

Final Review

Group 5

VidScribe

Intelligent Video-to-Text Summarization and Presentation Generator

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Introduction

- Video content is rapidly increasing in education, work and media.
- Manual review of long videos is slow and ineffective.
- Key insights are often hidden in lengthy recordings.
- An intelligent system can auto-summarize into text and slides.
- Combines audio + visuals for quick understanding.
- Improves learning, accessibility, and content sharing.

Objectives

- To Automate video content extraction and summarization to eliminate time-consuming manual review processes.
- To Enhance accessibility for users with hearing impairments and diverse learning needs through multi-modal content delivery.
- To Implement intelligent keyframe extraction using OpenCV motion analysis and optical flow techniques.

Objectives Contd...

- Generate comprehensive summaries by combining visual descriptions with audio transcripts using LLaMA Vision and Whisper models.
- Create structured PowerPoint presentations automatically from processed video content using python-pptx library.
- Support educational platforms with efficient video-to-text conversion and organized content delivery.

Problem Statement

- Video platforms produce massive amounts of educational and professional content that is difficult to consume fully.
- Manual reviewing of long videos is time-consuming, repetitive, and often leads to information loss.
- Important knowledge is buried inside hours of video without easy ways to extract or highlight it.
- Learners and professionals lack structured summaries that provide quick understanding of video material.
- Existing systems rarely combine audio, visuals, and text into a unified, easy-to-use presentation format.

Proposed System

- Extract keyframes from video using OpenCV (motion analysis & histogram comparison).
- Separate audio with MoviePy and transcribe using Whisper.
- Generate captions for keyframes via LLaMA Vision.
- Summarize captions and transcripts using LLaMA language model.
- Compile summarized text and visuals into a PowerPoint with python-pptx.
- Converts long videos into concise, presentation-ready content.

Tools and Approaches

1. Video Processing Tool: OpenCV

Function: Extracts keyframes from video using motion detection and histogram comparison.

Purpose: Identifies meaningful scene changes to select representative frames for summarization.

2. Audio Extraction Tool: MoviePy

Function: Separates audio from video files.

Purpose: Provides raw audio input required for speech transcription.

Tools and Approaches Contd...

3. Speech-to-Text (STT) Tool: Whisper (OpenAI)

Function: Converts video audio into accurate text transcripts using deep learning.

Purpose: Generates reliable textual content that forms the base for video summarization.

4. Vision Captioning Tool: LLaMA Vision

Function: Generates descriptive captions for extracted keyframes from the video.

Purpose: Adds visual context to the summary by understanding image content.

Tools and Approaches Contd...

5. Language Model Tool: LLaMA / GPT Model

Function: Combines transcripts and visual captions to produce concise summaries.

Purpose: Generates meaningful natural language summaries using NLP.

6. Presentation Generation Tool: python-pptx

Function: Automatically converts summarized content and keyframes into PowerPoint slides.

Purpose: Delivers final structured presentation output for easy viewing and sharing.

Tools and Approaches Contd...

7. Workflow Architecture: Sequential Processing Pipeline

Sequence: Video Input → OpenCV (Keyframe Extraction) → MoviePy (Audio Split) → Whisper (STT) → LLaMA Vision (Image Captions) → LLaMA/GPT (Text Summarization) → python-pptx (Slide Generation)

Purpose: Ensures modular and organized data flow for automated video summarization.

8. Backend Framework: Python

Function: Serves as the main programming language to integrate video, audio, and NLP modules.

Purpose: Manages the system logic and modular pipeline for smooth functionality.

Tools and Approaches Contd...

9. Deployment Option: Web or Desktop Interface

Function: Provides a simple application interface for users to upload video and generate outputs.

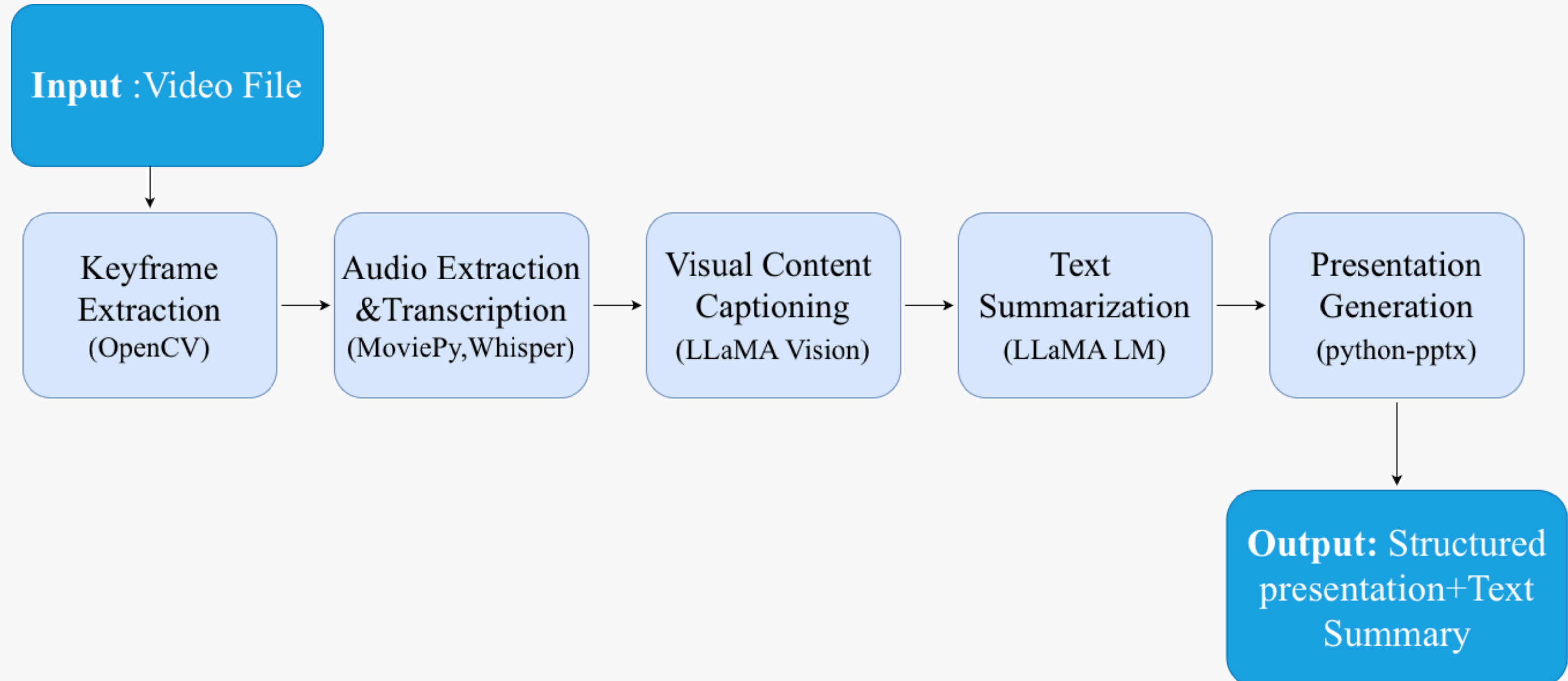
Purpose: Improves accessibility and usability across different platforms.

10. Storage Tool: Local File System / Cloud (optional)

Function: Stores video inputs, transcripts, and generated PPT files.

Purpose: Ensures data accessibility and enables reuse or download of previous results.

Flow of the Work



Working Principle

1. Input & Video Processing

- **Input:** The process begins when the user provides an Input Video File.
- **Keyframe Extraction:** The video file is processed using OpenCV (specifically, Optical Flow and Histogram analysis).
- **Principle:** This module detects significant motion or scene changes to select a minimum set of representative keyframes, which are crucial for summarizing the visual content without redundancy.

2. Audio Extraction & Transcription

- **Extraction:** The audio track is separated from the video using MoviePy.
- **Transcription:** The audio track is then passed to the Whisper (OpenAI) model.
- **Principle:** Whisper converts the spoken content into an accurate text transcript. This ensures the summary captures all verbal information provided in the video.

Working Principle Contd...

3. Visual Content Captioning

- Processing: The extracted keyframes are sent to the LLaMA Vision model.
- Principle: LLaMA Vision analyzes the images and generates natural language captions for each keyframe. These captions provide the necessary context for the visual elements, enabling a multimodal understanding of the content.

4. Text Summarization (LLM Core)

- Input Fusion: The core language model, LLaMA LM, receives both the audio transcript (from Whisper) and the keyframe captions (from LLaMA Vision).
- Principle: The LLaMA model combines and processes this multimodal information to produce a concise, coherent summary of the video content. This summary forms the textual backbone of the final presentation.

Working Principle Contd...

5. Presentation Generation

- Tool: The resulting summary text and selected key visuals are processed using the python-pptx library.
- Principle: This library automatically structures the summarized content and visuals into a standard, ready-to-use PowerPoint presentation. This eliminates the manual task of creating slides from long video content.

6. Output

- Final Delivery: The system generates the Structured Presentation (.pptx) and the raw Text Summary as the final output.
- Goal: The complete process converts lengthy videos into concise, structured content, enhancing accessibility and learning efficiency.

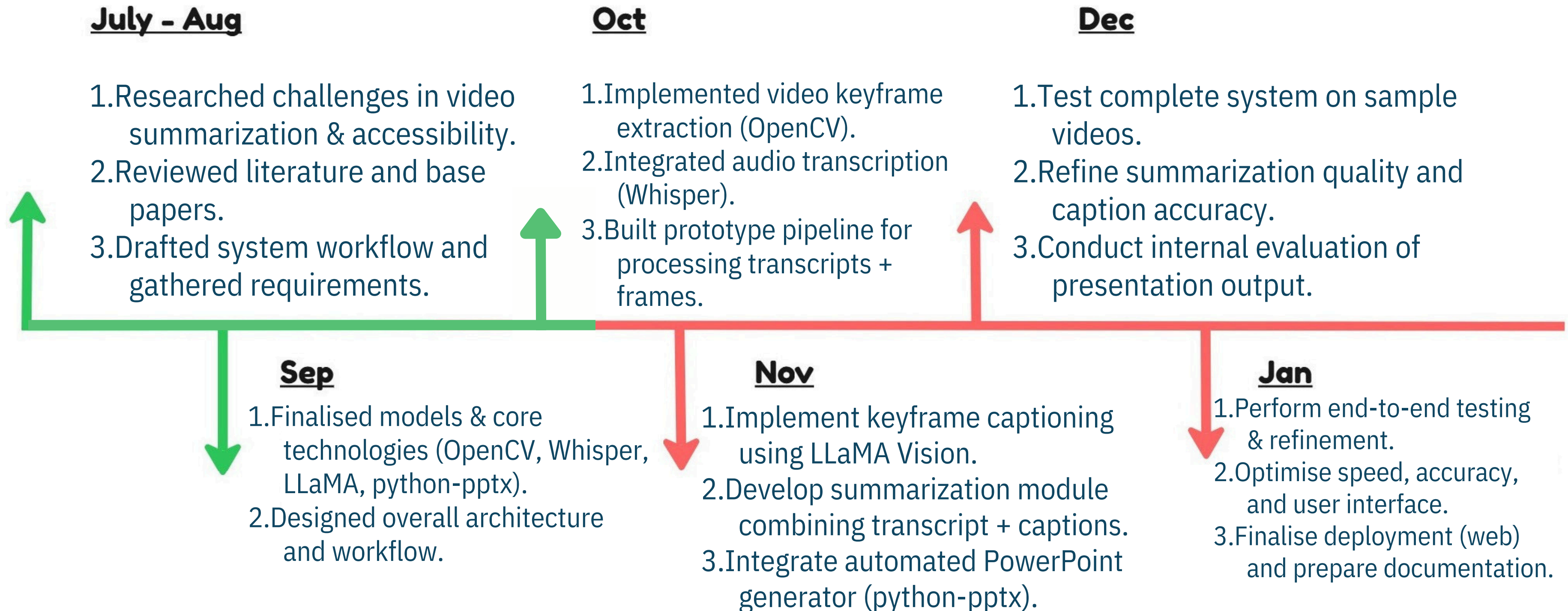
Work Done So Far

- Defined problem and objectives: converting long videos into concise, presentation-ready content.
- Selected deep learning and NLP methods: OpenCV, Whisper, and LLaMA models.
- Designed video-to-text workflow including keyframe extraction, transcription, captioning, summarization, and presentation.
- Tested OpenCV for keyframe extraction.
- Completed initial audio transcription with Whisper.
- Built prototype pipeline for processing transcripts +frames.

Work Plan Ahead

- Implement caption generation for keyframes using LLaMA Vision.
- Integrate summarization model to combine transcripts and captions.
- Develop automated PowerPoint generation module with python-pptx.
- Test the complete system on sample videos for accuracy and presentation quality.
- Refine modules for better performance and reliability.

Time Plan



Conclusion

- Provides intelligent video-to-text summarization.
- Converts lengthy videos into concise, structured content.
- Enhances accessibility and learning efficiency.
- Support educational platforms.

References

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THANK YOU

DO YOU HAVE ANY QUESTION ?